Correcting Aquaculture Facility-Induced Spectral Distortions for Improved Satellite Water Quality Retrieval in Marine Ranching Areas

Shengqiang Wang, Jiayu Meng, Deyong Sun, Xin Zhang, Zishen Li, Xiumei Zhang, Shuyan Lang, and Yongjun Jia

***Abstract*— Marine ranching plays a vital role in sustainable fishery resource utilization and marine ecosystem protection. The accurate water quality monitoring through remote sensing is essential in these areas; however, aquaculture facilities such as floating buoys introduce additional reflected signals that can distort remote sensing reflectance [$R_{rs}(\lambda)$], leading to significant errors in water quality parameter retrievals. Despite this challenge, previous studies have neither systematically quantified this interference nor developed effective correction methods. This study proposes a novel correction method to mitigate aquaculture-induced distortions in $R_{rs}(\lambda)$, enhancing the accuracy of remote sensing-based water quality assessments. Using the mussel aquaculture ranching area off Gouqi Island, China, as a case study, we systematically analyze the spectral influence of aquaculture facilities on $R_{rs}(\lambda)$ derived from Landsat observations. Comparative spectral analysis between affected and unaffected areas reveals that the high-reflectance characteristics of aquaculture facilities cause abnormally elevated $R_{rs}(\lambda)$ values, particularly in the shortwave infrared (SWIR) bands. To address this issue, we introduce an aquaculture facility influence factor and develop a pixel-based dynamic correction approach that adjusts for varying degrees of aquaculture-induced distortions across different pixels. Validations using field-measured $R_{rs}(\lambda)$ demonstrates that the mean absolute percentage errors (MAPEs) at wavelengths of 443, 483, 561, 655, 865, and 1609 nm decreased significantly from 17.6%, 19.8%, 14.8%, 26.8%, 50.9%, and 180.6% before correction to 9.5%, 11.0%, 9.2%, 10.4%, 7.7%, and 20.4%, respectively. The effectiveness of the correction method is further supported by improvements in the retrieval of water transparency (Z_{sd}). The sensitivity analysis further reveals that uncorrected Z_{sd} retrieval errors increase exponentially with aquaculture facility coverage, exceeding 60% when coverage reaches 10%, underscoring the necessity of correcting $R_{rs}(\lambda)$ distortions caused by aquaculture facilities. Overall, the proposed correction method provides a robust and adaptable framework for improving satellite-based water quality monitoring in complex aquaculture regions, with potential applicability across diverse marine ranching environments and integration with various satellite sensors.**

***Index Terms*— Aquaculture facility correction, marine ranching, remote sensing reflectance, satellite monitoring, water quality parameters.**

I. Introduction

MARITIME aquaculture plays a vital role in ensuring food security and fostering economic development [1]. However, the degradation of the ecological environment, exacerbated by human activities, has led to a decline in fishery resources [2], [3]. Marine ranching has emerged as a promising solution for the sustainable utilization of marine fishery resources and the protection of marine ecosystems [4]. This approach leverages modern engineering technologies and management strategies to restore habitats and support artificial breeding in suitable marine areas, yielding both economic and ecological benefits. It effectively balances increased fishery production, resource conservation, and environmental enhancement [5]. Furthermore, marine ranching can contribute to carbon sequestration, as marine organisms absorb and fix carbon dioxide, thus mitigating the greenhouse effect [6], [7].

As an integral component of the marine ecosystem, marine ranching must prioritize ecological considerations, especially in the context of its ongoing large-scale development. Monitoring water quality parameters is essential for evaluating environmental conditions and providing early warnings for potential ecological disasters. This is crucial for safeguarding the ecological security of marine ranching and ensuring their sustainable development toward green fisheries. Key water quality indicators, such as water transparency, chlorophyll a concentration, and total suspended particulate matter, reflect water clarity as well as eutrophication levels [8], [9]. These parameters also influence light utilization efficiency, which in turn affects primary productivity [10], [11]. Consequently,

Received 30 April 2025; revised 13 June 2025 and 15 June 2025; accepted 19 June 2025. Date of publication 25 June 2025; date of current version 3 July 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB3901304; and in part by the National Natural Science Foundation of China under Grant 42176181, Grant 42476173, and Grant 42176179. *(Corresponding author: Deyong Sun.)*

Shengqiang Wang and Deyong Sun are with the State Key Laboratory of Climate System Prediction and Risk Management and the School of Marine Sciences, Nanjing University of Information Science and Technology, Nanjing 210044, China, and also with the Key Laboratory of Space Ocean Remote Sensing and Application, Ministry of Natural Resources, Beijing 100081, China (e-mail: shengqiang.wang@nuist.edu.cn; sundeyong@nuist.edu.cn).

Jiayu Meng is with the School of Marine Sciences, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: 202212370015@nuist.edu.cn).

Xin Zhang and Zishen Li are with the Qilu Aerospace Information Research Institute, Jinan 250101, China (e-mail: zhangxin000181@aircas.ac.cn; lizishen@aircas.ac.cn).

Xiumei Zhang is with the Fisheries College, Zhejiang Ocean University, Zhoushan 316022, China (e-mail: xiumei1227@163.com).

Shuyan Lang and Yongjun Jia are with the National Satellite Ocean Application Service, Beijing 100081, China (e-mail: langshuyan@mail.nsoas.org.cn; jiayongjun@mail.nsoas.org.cn).

Digital Object Identifier 10.1109/TGRS.2025.3583085

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

effective real-time monitoring of water quality is essential for aligning marine environmental protection with the conservation of biological resources and for enhancing the intelligent management of marine ranching operations.

Traditionally, water quality parameters have been obtained through field surveys, laboratory analysis, or buoy observations. While these methods provide accurate data, they are often time-consuming and labor-intensive. More importantly, their spatial and temporal coverage is limited, making it difficult to meet the demands of real-time monitoring over large-scale marine environments. In contrast, water color remote sensing techniques have emerged as the primary method for acquiring large-scale spatial and temporal data on marine water quality parameters, thanks to their broad coverage and frequent revisit capabilities. Over the past decades, numerous water color satellite sensors have been launched, including solar-synchronous orbit sensors such as the Sea-Viewing Wide Field-of-View Sensor (SeaWiFS) and the Moderate Resolution Imaging Spectroradiometer (MODIS) for global marine water quality observations [12], as well as geostationary orbit sensors like the Geostationary Ocean Color Imager (GOCI) for local observations [13], [14]. Correspondingly, a variety of remote sensing inversion models have been developed for water quality assessment, including empirical regression methods [15], [16], [17], analytical/semianalytical algorithms [18], [19], [20], and machine learning techniques [21], [22]. These satellite sensor observations, combined with remote sensing algorithms, have facilitated the generation of various water color products, such as chlorophyll concentrations, transparency, and attenuation coefficients. These products provide valuable and reliable data for the dynamic monitoring of large-scale oceanic parameters, contributing significantly to the advancement of oceanography [23], [24]. However, the spatial resolution of existing ocean color remote sensing products is typically 1 km or 500 m. Such coarse resolution limits their ability to meet the demand for fine-scale monitoring of spatial–temporal variations in marine water quality parameters, particularly in small-scale marine ranching areas, which are typically only a few kilometers in size.

For smaller scale water bodies, such as estuaries, lakes, and rivers, researchers have increasingly turned to high-resolution land-based satellite sensors, such as Landsat series (30-m resolution) and the Sentinel series (20-m resolution), to remotely monitor water quality parameters, achieving notable progress [25], [26], [27], [28], [29]. For instance, Bonansea et al. [25] developed linear mixed models and used Landsat satellite data to successfully estimate water quality parameters in the Río Tercero reservoir. Lee et al. [26] employed Landsat-8 data to produce high-resolution Secchi disk depth (Z_{sd}) maps over an estuary near Xiamen, China, demonstrating that remotely sensed Z_{sd} closely aligned with in situ measurements, highlighting the potential of Landsat data for accurate water clarity estimation in estuaries, bays, and lakes. These studies suggest that high-resolution land-based satellite sensor data are more suitable and feasible for monitoring water quality parameters in marine ranching areas compared with traditional ocean color remote sensing data.

However, it is crucial to note that existing studies focus primarily on open-water bodies, where there are no ranch aquaculture facilities. Marine ranching areas, however, often contain aquaculture facilities, such as fish cages and floating rafts, which reflect sunlight and contribute additional reflected signals. These reflected signals, combined with water-leaving radiance, enter the satellite sensor's field of view, leading to deviations in the observed remote sensing reflectance [$R_{rs}(\lambda)$]. Since $R_{rs}(\lambda)$ forms the basis for inverting water quality parameters, such errors inevitably compromise the accuracy of these inversions. Therefore, quantifying the influence of aquaculture facilities on $R_{rs}(\lambda)$ and developing methods to correct for this impact is critical. This is essential to ensure the accuracy of water quality parameter inversions in marine ranching areas. Addressing this issue is essential to ensure the accuracy of water quality parameter inversions in marine ranching areas. However, up to date, no research has specifically addressed this challenge, representing a significant knowledge gap.

To address this knowledge gap, this study uses the mussel aquaculture ranching area of Gouqi Island, Shengsi, Zhejiang, as a case study to analyze the spectral influence of aquaculture facilities on $R_{rs}(\lambda)$ and propose a corresponding correction method. Based on this analysis, we develop a region-specific remote sensing algorithm for water transparency Z_{sd}, a key indicator of water quality. The corrected $R_{rs}(\lambda)$ data were then applied to retrieve Z_{sd} products, enabling a comprehensive evaluation of the correction method's effectiveness. The objective of this study is to address the challenges of accurate remote sensing inversion of water quality parameters in complex marine ranching environments by introducing a novel pixel-based dynamic correction method to effectively mitigate aquaculture facility-induced spectral distortions in $R_{rs}(\lambda)$. This approach not only enhances the accuracy and reliability of satellite-derived water quality assessments, such as water transparency, but also provides a valuable technical foundation for supporting more precise environmental evaluation and risk management in aquaculture-impacted coastal regions.

II. Data and Materials

A. Study Area and Sampling

Gouqi Island, located within Zhoushan Archipelago at the confluence of the Yangtze River and East China Sea, is characterized by nutrient-rich waters, favorable temperatures, and abundant feed organisms. As one of China's largest mussel-producing regions and a key national marine ranching demonstration zone [30], [31], it serves as an ecologically representative site for large-scale aquaculture environments. Moreover, Gouqi Island was selected for this study due to the availability of a comprehensive in situ dataset, which was acquired through four seasonal research cruises conducted within the mussel ranching area and adjacent marine zones. Specifically, in situ data were collected during cruises in November 2021 (autumn), January 2022 (winter), early June 2022 (late spring), and August 2022 (summer), providing robust support for the development and validation of our methods and ensuring the seasonal representativeness of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

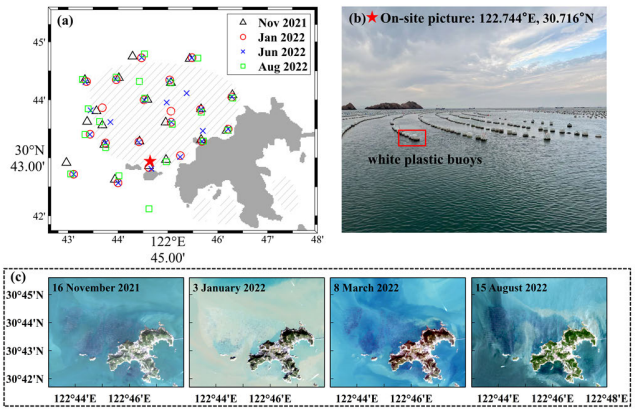


Fig. 1. (a) Locations of sampling stations of each cruise in the study region. (b) Photograph of the mussel aquaculture ranching area captured during field observations, with white plastic buoys labeled using the red rectangle. (c) True-colored composite imagery of the study area obtained from Landsat 8 OLI (bands 4, 3, and 2). Diagonal lines denote the mussel ranching area.

dataset. The study area specifically focused on the mussel aquaculture ranching area and the adjacent marine region of Gouqi Island, spanning from 30.695°N to 30.760°N latitude and 122.717°E to 122.800°E longitude. The primary in situ data used in this study include Secchi disk depth Z_{sd}, water remote sensing reflectance, $R_{rs}(\lambda)$, and remote sensing reflectance of aquaculture facility [$R_{rs_buoy}(\lambda)$].

The "off-bottom" long-line culture method is employed in the mussel ranching area of Gouqi Island [see Fig. 1(b)], where white plastic buoys are used to suspend mussel-growing lines that are anchored to the seabed. Each buoy is approximately 30 cm in diameter, and several buoys are connected in series to form raft-like rows. The spacing between adjacent rows is around 3–4 m, and buoys within each row are spaced approximately 1 m apart. Once mussels reach harvest size, they are collected without replacing the buoys. As a result, there is little seasonal variation in the aquaculture structure itself. Considering the spatial resolution of commonly used satellite sensors, such as Landsat 8 Operational Land Imager (OLI; 30 m), a single pixel typically contains a large number of buoys, which inevitably influences the reflected signal received by satellites, affecting the accuracy of remote sensing observations.

B. In Situ Data

$R_{rs}(\lambda)$ was measured within the spectral range of 350–2500 nm using an ASD FieldSpec[1] Pro FR spectroradiometer with a spectral increment of 1 nm. To minimize the influence of ship shadowing, the measurement azimuth and zenith angles were set to 135° and 40°, respectively [32]. Measurements were conducted under stable and appropriate sunlight conditions, typically between 10:00 A.M. and 3:00 P.M. Radiance spectra were collected for the reference panel, water, and skylight, with ten spectra acquired for each target. After excluding anomalous spectra, the remaining spectra were averaged to obtain the final reflectance spectrum for each target. The $R_{rs}(\lambda)$ spectra were calculated using the following equation [32]:

$$R_{rs}(\lambda) = \frac{L_{sw}(\lambda) - \rho L_{sky}(\lambda)}{\pi L_p(\lambda)/\rho_p} - \Delta \tag{1}$$

where $L_{sw}(\lambda)$, $L_{sky}(\lambda)$, and $L_p(\lambda)$ represent the averaged radiance spectra of the water, sky, and reference gray panel, respectively; ρ_p denotes the diffuse reflectance of the reference gray panel provided by the manufacturer; ρ indicates the Fresnel reflectance of sea water; and Δ represents a lump sum of the "residual" sky glint and sun glint. The values of ρ and Δ are determined following the method described by Lee et al. [32], which is specifically designed to correct for surface-reflected skylight in above-water $R_{rs}(\lambda)$ measurements.

The remote sensing reflectance of aquaculture facilities (buoys), $R_{rs_buoy}(\lambda)$, was measured in a similar manner but at close range, ensuring that the sensor's field of view was fully occupied by the buoy surface. This setup effectively eliminated interference from water-surface reflected sky light. Therefore, sky radiance subtraction was not applied, and $R_{rs_buoy}(\lambda)$ was calculated using the following equation:

$$R_{rs_buoy}(\lambda) = \frac{L_b(\lambda)}{\pi L_p(\lambda)/\rho_p} \tag{2}$$

where $L_b(\lambda)$ and $L_p(\lambda)$ represent the radiance of the aquaculture facilities and the reference gray panel, respectively. As before, ρ_p denotes the diffuse reflectance of the reference gray panel provided by the manufacturer.

The collected $R_{rs}(\lambda)$ and $R_{rs_buoy}(\lambda)$ data were resampled to Landsat 8 OLI spectral bands using response functions from the U.S. Geological Survey (USGS: earthexplorer.usgs.gov), as expressed in the following equation:

$$R_{rs}(\lambda_k) = \frac{\int_{\lambda_i}^{\lambda_j} S(\lambda) R_{rs}(\lambda) d\lambda}{\int_{\lambda_i}^{\lambda_j} S(\lambda) d\lambda} \tag{3}$$

where λ_k is the central wavelength of Landsat 8 OLI bands, λ_i and λ_j represent the lower and upper bounds of the spectral band λ_k, respectively, and $S(\lambda)$ indicates the spectral response function. Based on the resampled data, we conducted a comparative analysis with satellite spectra to assess the impact of buoys, which informed the development of a correction algorithm. In addition, the resampled $R_{rs}(\lambda)$ data were utilized to develop a remote sensing model for water transparency inversion.

C. Satellite Data

The satellite data utilized in this study were obtained from the Landsat 8 OLI sensor, which provides data across seven spectral bands: 443, 483, 561, 655, 865, 1609, and 2201 nm. To match the timing of our field cruises, we selected and downloaded four Level-1 images that were visually confirmed to be cloud-free, each representing a different season: November 16, 2021 (autumn); January 3, 2022 (winter); March 24, 2022 (spring); and August 15, 2022 (summer). Atmospheric correction of the Landsat 8 OLI images was

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

performed using the ACOLITE software (version 20220222.0) with the dark spectrum fitting (DSF) algorithm [33], [34]. The DSF algorithm, developed for sensors with meter-level resolution, has proven effective in turbid water environments.

D. Accuracy Assessment

To assess the performance of the remote sensing inversion model and the correction method, we employed four accuracy metrics, including the coefficient of determination (R^2), mean absolute percentage error (MAPE), mean bias (Bias), and root mean square error (RMSE). The formulas for these metrics are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i' - y_i\right)^2}{\sum_{i=1}^{N}(y_i - \overline{y})^2} \tag{4}$$

$$\text{Bias} = \frac{1}{N}\sum_{i=1}^{N}(y\prime_i - y_i) \tag{5}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y\prime_i - y_i)^2} \tag{6}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y\prime_i - y_i}{y_i}\right| \times 100\% \tag{7}$$

where y_i and y_i' indicate the measured and estimated values of the ith sample, respectively, $\bar{y}$ denotes the mean value of measured values, and N is the number of samples.

III. Results

A. In Situ Measured Spectra of $R_{rs}(\lambda)$ and $R_{rs_Buoy}(\lambda)$

The in situ measured water remote sensing reflectance $R_{\text{rs}}(\lambda)$ data obtained in this study are presented in Fig. 2(a). The $R_{\text{rs}}(\lambda)$ values show notable variation across the visible light spectrum. Specifically, the $R_{\text{rs}}(\lambda)$ values are relatively low in the blue wavelength range but gradually increase as the wavelength increases, reaching a peak between 550 and 580 nm. For some samples, the peak exceeds 0.02 sr^{-1}, followed by a steep decline between 600 and 700 nm. In the near-infrared (NIR) range, rather than continuing to decrease, the data exhibit a secondary peak around 810 nm, due to strong backscattering from suspended particulate matter [35]. These spectral curves demonstrate characteristics that are both similar in magnitude and shape to the typical spectral features of coastal waters [35], [36].

The remote sensing reflectance of ranch aquaculture facilities $R_{\text{rs_buoy}}(\lambda)$ exhibits significant differences from that of water bodies, both in spectral shape and magnitude [see Fig. 2(b)]. Notably, $R_{\text{rs_buoy}}(\lambda)$ shows much higher values, particularly in NIR and shortwave infrared (SWIR) regions. The spectral values of $R_{\text{rs_buoy}}(\lambda)$ increase continuously up to 500 nm, after which they fluctuate slightly, ranging from 0.2 and 0.3 sr^{-1}. In contrast, the water remote sensing reflectance, $R_{\text{rs}}(\lambda)$, remains below 0.05 sr^{-1} throughout [see Fig. 2(a)]. Even in the SWIR region, $R_{\text{rs_buoy}}(\lambda)$ stays around 0.2 sr^{-1}, while $R_{\text{rs}}(\lambda)$ is known to approach 0 due to the strong absorption by water. The high R_{rs}_buoy(λ) values observed in the NIR and SWIR bands are primarily attributed

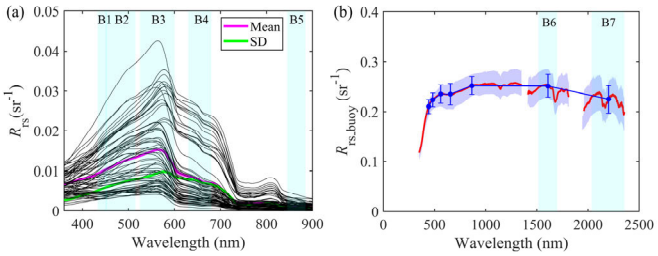


Fig. 2. (a) In situ hyperspectral measurements of water remote sensing reflectance, $R_{\text{rs}}(\lambda)$ and (b) remote sensing reflectance of ranch aquaculture facilities (buoy structures), $R_{\text{rs_buoy}}(\lambda)$, as well as the resampled $R_{\text{rs_buoy}}(\lambda)$ spectra aligned with the spectral bands of Landsat 8 OLI (represented by the blue line with error bars). Cyan-colored bars represent the spectral ranges of the seven Landsat 8 OLI bands: B1 (433–453 nm), B2 (450–515 nm), B3 (525–600 nm), B4 (630–680 nm), B5 (845–885 nm), B6 (1515–1697 nm), and B7 (2037–2355 nm), according to Irons et al. [37].

to the physical and optical properties of the buoys. These buoys are typically made of white plastic materials, such as expanded polystyrene (EPS) or polyvinyl chloride (PVC), which have low absorption and high-diffuse reflectance in these spectral ranges. Their smooth or semimatte surfaces further enhance diffuse reflection, resulting in consistently high and spectrally stable reflectance across the NIR–SWIR range, which manifests as a plateau in the observed spectra.

B. Impact of Ranch Aquaculture Facilities on $R_{rs}(\lambda)$ Spectra

The high-reflectance characteristics of ranch aquaculture buoys, inevitably affect the water's remote sensing reflectance received by satellite sensors, which further introduces errors in the remote sensing inversion of water quality parameters. To assess the spectral impact of aquaculture buoys on water $R_{\text{rs}}(\lambda)$, we divided the satellite image data into two regions based on the boundary of the mussel aquaculture ranching area: data within the mussel ranching area, where aquaculture facilities may influence the reflectance, and data outside the mussel ranching zone, unaffected by such influence. The boundary was delineated primarily using official administrative datasets and further refined through satellite image interpretation and field survey records. As shown in Fig. 3(a)–(d), the average spectrum outside the ranching area exhibits typical spectral features of turbid water, characterized by a peak in the green band and nonzero $R_{\text{rs}}(865)$. In the SWIR region, the values of $R_{\text{rs}}(1609)$ and $R_{\text{rs}}(2201)$ approach 0 due to strong water absorption. These spectral features are consistently observed across all four satellite images, confirming their presence. In contrast, the average spectrum within the ranching area shows notable differences, particularly in $R_{\text{rs}}(1609)$ and $R_{\text{rs}}(2201)$, where the values are around 0.003 sr^{-1}, substantially higher than the average values observed in the open waters outside the mussel ranching area. This characteristic is also consistently observed in all four seasonal images. The elevated reflectance values are attributed to the presence of the buoys, as evidenced by the significantly higher $R_{\text{rs_buoy}}(\lambda)$ values in the NIR bands, as shown in Fig. 2(b).

Furthermore, we conducted a statistical analysis of $R_{\text{rs}}(2201)$ for all pixels both inside and outside the mussel ranching area and found distinct differences in their

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

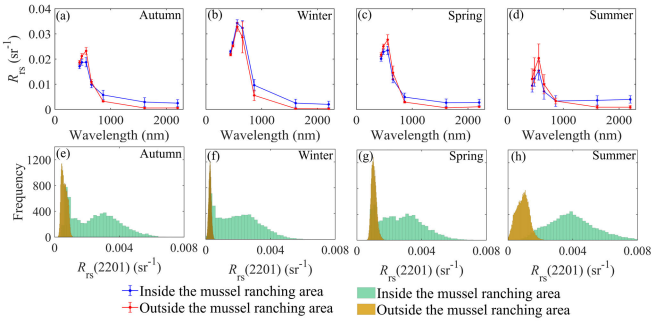


Fig. 3. Mean remote sensing reflectance spectra measured within (affected by aquaculture facilities) and outside (unaffected by aquaculture facilities) mussel aquaculture ranching area for four seasons. (a) November 2021 (autumn). (b) January 2022 (winter). (c) March 2022 (spring). (d) August 2022 (summer). Histograms of $R_{rs}(2201)$ for all pixels within and outside the mussel ranching area for each corresponding period: (e) November 2021, (f) January 2022, (g) March 2022, and (h) August 2022.

distribution [see Fig. 3(e)–(h)]. As anticipated, the $R_{rs}(2201)$ values outside the ranching area were consistently concentrated in the lower value range across all four seasons. During the autumn and winter [see Fig. 3(e) and (f)], $R_{rs}(2201)$ values were generally below 0.001 sr^{-1}. Although $R_{rs}(2201)$ values increased in the spring and summer, the average remained around 0.001 sr^{-1}, with the majority of maximum values also falling below 0.002 sr^{-1} [see Fig. 3(g) and (h)]. In contrast, pixels inside the mussel ranching area exhibited a wider distribution of $R_{rs}(2201)$ values. Notably, these values were substantially higher than those observed outside the ranching area, with average values ranging between 0.003 and 0.004 sr^{-1} and maximum values reaching approximately 0.008 sr^{-1}. These statistical results further underscore the significant impact of the aquaculture facilities on $R_{rs}(\lambda)$.

C. Development of the Correct Method for $R_{rs}(\lambda)$

It is important to note that the aquaculture buoys are in white [see Fig. 1(b)], indicating that their impact may not be limited to $R_{rs}(1609)$ and $R_{rs}(2201)$ but may also extend to the visible light bands, thereby affecting the inversion of water color parameters. Therefore, correcting the impact of aquaculture facilities on $R_{rs}(\lambda)$ is necessary. Since the buoys are not uniformly distributed across the entire sea surface, their influence on $R_{rs}(\lambda)$ of each pixel is not constant but varies depending on the extent of buoy coverage within the satellite's field of view. To address this variability, we propose a dynamic, pixel-based adaptive correction method that estimates and compensates for the buoy contribution on a per-pixel basis. The correction method is defined by the following equation:

$$R_{rs_corrected}(\lambda) = R_{rs}(\lambda) - f \times R_{rs_buoy}(\lambda) \tag{8}$$

where $R_{rs}(\lambda)$ and $R_{rs_corrected}(\lambda)$ represent the remote sensing reflectance of pixels before and after corrections for the impact of aquaculture facilities, respectively, and $R_{rs_buoy}(\lambda)$ refers to the endmember spectrum of the remote sensing reflectance for the aquaculture facilities, which was obtained from field measurements during the cruise. The factor f represents the influencing factor of aquaculture facilities, and we assume that f is independent of wavelength but varies with the buoy coverage within each pixel. It is worth noting that our correction model [see (8)] is based on the widely used and demonstrably effective linear spectral mixing framework [38], [39], which models the satellite-observed reflectance as a linear combination of water and buoy reflectance. Within this framework, the factor f represents the fraction of the pixel area covered by aquaculture buoys projected in nadir view. Since f reflects a geometric coverage proportion rather than a spectrally varying property, assuming that f to be spectrally invariant is considered reasonable.

The calculation of the factor f is a critical step during the $R_{rs}(\lambda)$ correction process. We assume that the abnormally high $R_{rs}(1609)$ and $R_{rs}(2201)$ observed within the mussel aquaculture ranching area are primarily due to the high-reflectance characteristics of the aquaculture buoys in these bands, thus providing a basis for calculating f. However, it is important to note that $R_{rs}(1609)$ and $R_{rs}(2201)$ values outside the ranching area, as obtained from Landsat 8 OLI, are not necessarily 0 (see Fig. 3). This may be attributed to contributions from highly turbid waters or systematic biases introduced during the preprocessing of satellite remote sensing data. Therefore, we propose a method to calculate f, considering both the spectral value differences inside and outside the mussel aquaculture ranching area, as well as potential systematic biases. Specifically, the factor f is calculated as the ratio of each pixel's deviation in $R_{rs}(2201)$ from the background value (representing areas without aquaculture facility influence) to the aquaculture buoy reflectance $R_{rs_buoy}(2201)$, as described by the following equation:

$$f = \frac{R_{rs_in}(2201) - b}{R_{rs_buoy}(2201)} \tag{9}$$

where $R_{rs_in}(2201)$ represents the $R_{rs}(2201)$ value of the pixel inside the mussel aquaculture ranching area; $R_{rs_buoy}(2201)$ refers to the remote sensing reflectance of aquaculture buoys at 2201 nm; and b represents the background $R_{rs}(2201)$ value of a pixel unaffected by aquaculture buoys (i.e., a pixel outside the ranching zone), accounting for the contributions from highly turbid waters or systematic bias. Based on the histogram distribution of $R_{rs}(2201)$ values outside the ranching area, we define b as the mean $R_{rs}(2201)$ plus two standard deviations (SDs) for all pixels outside the ranching area, considering this value as the stable upper limit of $R_{rs}(2201)$ unaffected by the aquaculture buoys. The equations for calculating b are shown as follows:

$$b = \text{mean}\left(R_{rs_out}(2201)\right) + 2a \tag{10}$$

where mean[$R_{rs}(2201)$] and a represent the mean and SD of $R_{rs}(2201)$ selected from the area outside the ranching zone, respectively. Based on these calculations, the value of f for pixels affected by aquaculture facilities ranges between 0 and 1. If f is negative, it indicates that the pixel is not influenced by the aquaculture facilities and therefore requires no correction. This dynamic correction accounts for the spatially varying impact of aquaculture facilities and allows pixel-level adjustment.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

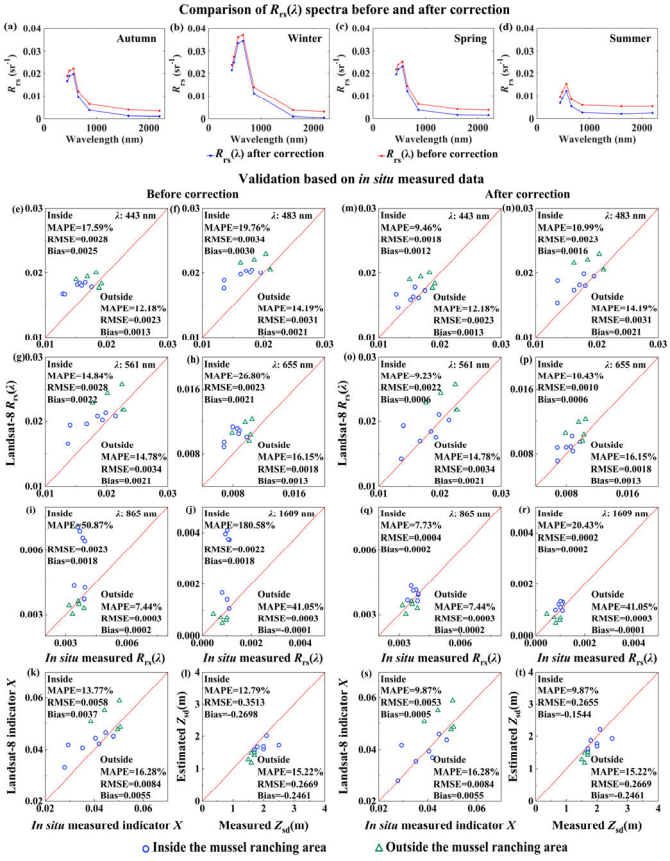


Fig. 4. Comparison of $R_{rs}(\lambda)$ spectra before and after corrections for the influence of aquaculture facilities for (a) November 2021, (b) January 2022, (c) March 2022, and (d) August 2022. (e)–(j) Validation of uncorrected $R_{rs}(\lambda)$ at wavelengths of 443, 483, 561, 655, 865, and 1609 nm, respectively. (k) Spectral indicator X and (l) satellite-derived Z_{sd} based on in situ measurements. (m)–(r) Corresponding validations after correction for $R_{rs}(\lambda)$, followed by (s) spectral indicator X and (t) satellite-derived Z_{sd}. Blue circles represent samples located within the mussel aquaculture ranching area where correction was applied, while green triangles denote samples outside the ranching zone that did not require correction.

It is important to clarify that both the 2201 and 1609 nm bands are characterized by extremely low water-leaving reflectance due to strong water absorption, making them suitable candidates for use as reference bands in estimating f. We ultimately selected the 2201 nm band because it lies at a longer wavelength, where water absorption is even stronger and the reflectance of natural water is closer to 0 compared with the 1609 nm band. This results in greater spectral stability and reduced sensitivity to water variability, thereby providing a more robust basis for isolating aquaculture facility-induced reflectance spectral distortions.

To illustrate the variations in $R_{rs}(\lambda)$ spectra before and after correcting for the impact of the aquaculture facilities, we selected four representative pixel samples from Landsat images spanning different seasons: November 16, 2021, January 3, 2022, March 8, 2022, and August 15, 2022. As shown in Fig. 4(a)–(d), it is clear that, following the correction, the $R_{rs}(1609)$ and $R_{rs}(2201)$ values no longer display abnormally high readings, and the corrected $R_{rs}(\lambda)$ spectra now exhibit the typical characteristics of normal water bodies.

Furthermore, we validated the corrected $R_{rs}(\lambda)$ data using in situ measurements collected during our cruise observations. Satellite-derived $R_{rs}(\lambda)$ values were matched to field observations within a $\pm$12-h temporal window and a 3 $\times$ 3 pixel spatial window centered on each in situ measurement location. The proposed correction method was subsequently applied to mitigate the influence of aquaculture facilities, and the average corrected $R_{rs}(\lambda)$ within this spatial window served as the representative value for comparison with the corresponding in situ measurement. In total, 12 matchup pairs between satellite-derived and field-measured data were obtained: five located in waters outside the mussel ranching area and seven within it. It should be noted that although a $\pm$12-h window is commonly used in satellite–in situ matchup studies, it may introduce temporal mismatches in dynamic coastal waters. However, due to Landsat 8's 16-day revisit cycle and frequent cloud cover, using a narrower window would greatly reduce the number of valid matchups. Thus, we adopted the $\pm$12-h window as a practical compromise to balance temporal accuracy and data availability.

Validation results indicate that in waters outside the ranching area where no correction for aquaculture facilities was necessary [see Fig. 4(e)–(j)], the satellite $R_{rs}(\lambda)$ values exhibited good reliability, with MAPE ranging from 12.2% to 16.2% in visible bands. In contrast, substantial deviations in $R_{rs}(\lambda)$ were observed within the ranching area before correction (see Fig. 4(e)–(j) and Table I), clearly indicating spectral distortion caused by the aquaculture facilities. After applying the correction method (see Fig. 4(m)–(r) and Table I), the corrected $R_{rs}(\lambda)$ values within the ranching area aligned more closely with the 1:1 reference line, indicating significant improvements in spectral accuracy. Specifically, MAPE values at wavelengths of 443, 483, 561, 655, 865, and 1609 nm decreased significantly from 17.6%, 19.8%, 14.8%, 26.8%, 50.9%, and 180.6% to 9.5%, 11.0%, 9.2%, 10.4%, 7.7%, and 20.4%, respectively. Corresponding reductions in RMSE values were also observed across these bands, with an average decrease of approximately 0.001 sr^{-1}. These results demonstrate that the proposed correction method substantially improves the consistency between satellite-derived $R_{rs}(\lambda)$ data and in situ measurements in aquatic environments affected by aquaculture facilities, thereby enhancing the reliability of remote sensing applications in marine ranching environments.

D. Retrieval of Water Transparency From the Corrected $R_{rs}(\lambda)$

To evaluate the effectiveness of the correction method developed in this study, we employed the corrected $R_{rs}(\lambda)$ to derive water transparency Z_{sd} in the mussel aquaculture ranching area and the adjacent sea region. A region-specific remote sensing algorithm was developed based on field-measured match-ups between $R_{rs}(\lambda)$ and Z_{sd} data. To obtain the optimal remote sensing algorithm for Z_{sd} estimation, we designed eight spectral indices to fully exploit the spectral characteristics of $R_{rs}(\lambda)$. Each spectral index was assessed using all possible combinations of Landsat 8 OLI bands, and the relationships between these indices and Z_{sd} were analyzed under different band combinations. The optimal band

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF RMSE, MAPE, AND BIAS VALUES BEFORE AND AFTER CORRECTIONS

Variable	Accuracy metrics	Before correction	After correction
$R_{rs}(443)$	MAPE (%)	17.59%	9.46%
	RMSE (sr^{-1})	0.0028	0.0018
	Bias (sr^{-1})	0.0025	0.0012
$R_{rs}(483)$	MAPE (%)	19.76%	10.99%
	RMSE (sr^{-1})	0.0034	0.0023
	Bias (sr^{-1})	0.0030	0.0016
$R_{rs}(561)$	MAPE (%)	14.84%	9.23%
	RMSE (sr^{-1})	0.0028	0.0022
	Bias (sr^{-1})	0.0022	0.0006
$R_{rs}(655)$	MAPE (%)	26.80%	10.43%
	RMSE (sr^{-1})	0.0023	0.0010
	Bias (sr^{-1})	0.0021	0.0006
$R_{rs}(865)$	MAPE (%)	50.87%	7.73%
	RMSE (sr^{-1})	0.0023	0.0004
	Bias (sr^{-1})	0.0018	0.0002
$R_{rs}(1609)$	MAPE (%)	180.58%	20.43%
	RMSE (sr^{-1})	0.0022	0.0002
	Bias (sr^{-1})	0.0018	0.0002
indicator X	MAPE (%)	13.77%	9.87%
	RMSE (sr^{-1})	0.0058	0.0053
	Bias (sr^{-1})	0.0037	0.0005
Z_{sd}	MAPE (%)	12.79%	9.87%
	RMSE (m)	0.3513	0.2655
	Bias (m)	-0.2698	-0.1544

combination for each spectral index was subsequently identified. Furthermore, various functional forms were compared for each index to establish the most accurate Z_{sd} retrieval model. The model performance was rigorously evaluated using leave-one-out cross validation, with R^2, MAPE, and RMSE serving as the evaluation metrics. Based on these accuracy metrics, the optimal remote sensing algorithm for Z_{sd} estimation was determined, as represented in the following equation:

$$Z_{sd} = 194.13X^2 - 50.39X + 3.47 \tag{11}$$

where X denotes $[R_{rs}(443) + R_{rs}(561)]/[R_{rs}(443)/R_{rs}(561)]$.

The model's fitting performance is illustrated in Fig. 5(a), where most sample points are closely aligned with the fit curve. Fig. 5(b) presents the comparison between the estimated and measured Z_{sd} values obtained through leave-one-out cross-validation. The estimated Z_{sd} values showed good agreement with observed measurements, yielding an RMSE of 0.54 m and an MAPE of 26.2%. These findings indicate that the proposed remote sensing approach effectively estimates Z_{sd} using $R_{rs}(\lambda)$ data from Landsat 8 OLI bands.

We applied the developed remote sensing algorithm to derive Z_{sd} from the corrected Landsat 8 OLI $R_{rs}(\lambda)$ data, both to evaluate the effectiveness of the correction method in mitigating the impact of aquaculture facilities and to investigate the spatial and temporal variations Z_{sd} in the mussel aquaculture ranching area. Before applying the algorithm, we validated the satellite-derived spectral index X and Z_{sd} using the satellite–ground matchup dataset as described in Section III-C. The proposed correction method was first applied to minimize aquaculture-induced distortions in $R_{rs}(\lambda)$, after which the Z_{sd} model was employed to estimate Z_{sd} at corresponding

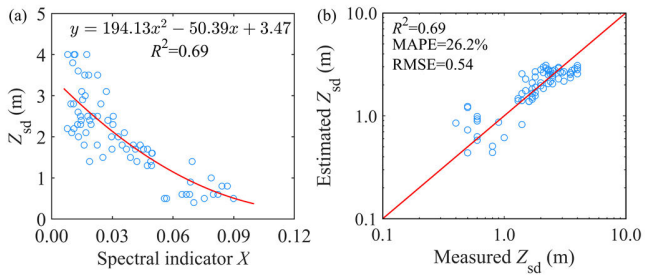


Fig. 5. (a) Scatter plot of Z_{sd} and the spectral indicator X derived from $R_{rs}(\lambda)$ and (b) comparison between Z_{sd} estimated by the model and the in situ measured values The red solid and dashed lines denote the fitting curve and the 1:1 line, respectively.

matching points. To further evaluate correction efficacy, the same Z_{sd} algorithm was applied to the uncorrected $R_{rs}(\lambda)$ data for comparative assessment against the corresponding in situ Z_{sd} measurements. The results indicate that the spectral index X derived from the corrected $R_{rs}(\lambda)$ exhibited strong agreement with in situ values, clustering closely around the 1:1 reference line, and yielding an MAPE of 9.87%, an RMSE of 0.0053, and a bias of 0.0005 (see Fig. 4(s) and Table I). This improved spectral index performance translated into more accurate Z_{sd} estimates, which also aligned well with field observations, producing an MAPE of 9.87%, an RMSE of 0.27 m, and a bias of −0.15 m (see Fig. 4(t) and Table I). In contrast, spectral index X values derived from uncorrected $R_{rs}(\lambda)$ data exhibited notable overestimation, characterized by a bias of 0.0037, higher MAPE (13.77%), and increased RMSE (0.0058) (see Fig. 4(k) and Table I). Correspondingly, the resulting Z_{sd} estimates from uncorrected data showed significant underestimation, reflected by a bias of –0.27 m, higher MAPE (12.79%), and greater RMSE (0.35 m) (see Fig. 4(l) and Table I). These findings confirm that the proposed correction method effectively mitigates the spectral interference caused by aquaculture facilities, thereby substantially enhancing the accuracy and reliability of satellite-derived water quality parameters such as Z_{sd} in complex marine ranching environments.

The proposed correction method and the Z_{sd} remote sensing model were applied to Landsat 8 OLI satellite images, selecting one image from each of the four seasons to examine the spatial and temporal variations of Z_{sd} in the marine ranch and adjacent regions [see Fig. 6(a)–(d)]. Mean Z_{sd} values were also statistically compared inside and outside the marine ranching area [see Fig. 6(e)]. In autumn 2021 [see Fig. 6(a)], Z_{sd} was relatively high across both the marine ranch and nearby waters, with an average of 1.84 m within the ranch. This closely aligns with the average Z_{sd} of 1.86 m obtained from our in situ measurements within the ranch (see Table II). In winter (January 2022) [see Fig. 6(b)], Z_{sd} reached its lowest levels in the ranch and adjacent waters, with most areas falling below 1 m. The average Z_{sd} inside and outside the ranch was 0.63 and 0.62 m, respectively, consistent with our in situ measurements of 0.70 m (see Table II). The lower Z_{sd} values in winter are probably influenced by a strong wind event that occurred in early January 2022 in the study area, which can intensify

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

TABLE II

SUMMARY STATISTICS OF IN SITU MEASURED Z_{sd} INSIDE AND OUTSIDE THE MUSSEL RANCHING AREA COLLECTED DURING FOUR CRUISES. MAX: MAXIMUM, MIN: MINIMUM, SD: STANDARD DEVIATION, AND CV: COEFFICIENT OF VARIATION

Area	Date	Max (m)	Min (m)	Mean (m)	SD (m)	CV (%)
Inside the mussel ranching area	Nov. 2011	2.50	1.40	1.86	0.30	14.6
	Jan. 2022	1.00	0.40	0.70	0.20	27.0
	Jun. 2022	5.00	2.40	3.50	0.70	20.1
	Aug. 2022	3.10	2.10	2.52	0.30	11.5
Outside the mussel ranching area	Nov. 2011	1.70	1.40	1.58	0.10	6.7
	Jan. 2022	0.90	0.50	0.70	0.10	21.3
	Jun. 2022	4.00	2.10	2.90	0.70	23.1
	Aug. 2022	2.60	1.30	1.64	0.40	24.2

vertical mixing in the water column and resuspend bottom sediments, thereby reducing Z_{sd}. By spring (March 2022) [see Fig. 6(c)], the water transparency had increased in both the marine ranch and the adjacent sea areas. Water transparency had increased, with a typical Z_{sd} of around 1.02 m outside the ranch and an average of 1.45 m within the ranch. During summer 2022 [see Fig. 6(d)], most parts of the marine ranch and surrounding waters exhibited high water transparency, with average Z_{sd} values of 1.74 m within the ranch and 1.28 m in adjacent areas.

In addition to the seasonal characteristics, we found that Z_{sd} inside the mussel culture area was significantly higher than that outside. These differences may be attributed to mussel culture activities. Mussels are filter-feeding organisms capable of removing small plankton and other organic particles from the water [40], [41], [42], thus potentially increasing Z_{sd}. Moreover, the structural components of raft-based aquaculture, such as the rafts themselves and the suspension ropes, can act as physical barriers, reducing flow velocity by impeding tidal or current flows [43]. Raft aquaculture may also enhance biological sedimentation in the farming area [44], [45], [46]. These processes may collectively contribute to the higher Z_{sd} observed within the mussel culture area compared with areas outside.

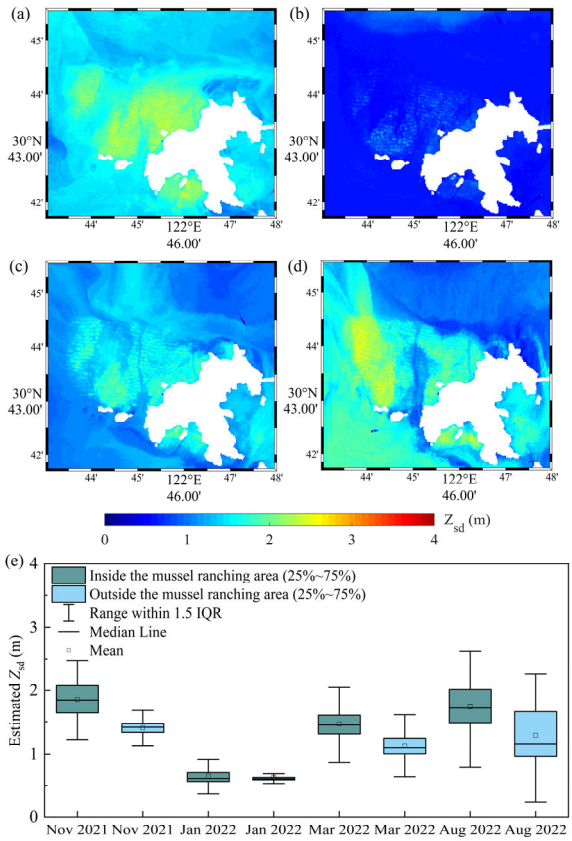


Fig. 6. Spatial distributions of satellite-derived Z_{sd} derived from Landsat 8 OLI imagery during four seasonal observation periods. (a) November 2021 (autumn). (b) January 2022 (winter). (c) March 2022 (spring). (d) August 2022 (summer). The accompanying statistical comparisons (e) present the distribution of Z_{sd} values within and outside the mussel aquaculture ranching area for each corresponding observation period.

IV. DISCUSSION

A. Necessity for Correcting Aquaculture Facility Impacts on Satellite $R_{rs}(\lambda)$

The high reflectance characteristics of aquaculture buoys result in abnormally high satellite observed $R_{rs}(\lambda)$ values, particularly in the SWIR bands (see Fig. 3). Notably, as shown in Fig. 3, these abnormally high values are not constant but exhibit significant variability. Some pixels within the aquaculture ranching area even show $R_{rs}(2201)$ values similar to those outside the ranching area. These observations indicate that the influence of aquaculture facilities on pixels within the mussel ranching area varies, with some pixels minimally affected or not affected at all. This occurs because relatively open water areas within the aquaculture ranching zone, as observed during field surveys, exceed the Landsat sensor's field of view, thereby making the facility-related influence on those pixels negligible. Therefore, accurately quantifying the degree of aquaculture facility influence on each pixel, in a dynamic manner, is critical for correcting $R_{rs}(\lambda)$ accurately. The correction method proposed in this study introduces an aquaculture facility influencing factor f and a pixel-based dynamic calculation approach to account for variability in the degree of influence from aquaculture facilities. However, questions remain regarding the significance of deviations in $R_{rs}(\lambda)$ caused by aquaculture facilities, the magnitude and variability of f, and whether correcting for this influence is always necessary. These are issues that warrant further investigation.

To address these issues, a statistical analysis was conducted of all $R_{rs}(\lambda)$ data at the 2201 nm band, comparing abnormally high values between pixels influenced by aquaculture facilities and those unaffected, as well as examining the variability of the influencing factor f [see Fig. 7(a) and (b)]. We found that abnormal high values induced by aquaculture facilities can reach up to 0.007 sr^{-1} at 2201 nm, with most values ranging from 0.001 to 0.004 sr^{-1} and an average of approximately 0.002 sr^{-1}. These discrepancies markedly exceed the $R_{rs}(2201)$ values observed in pixels outside the aquaculture ranching area, where $R_{rs}(2201)$ generally remains

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

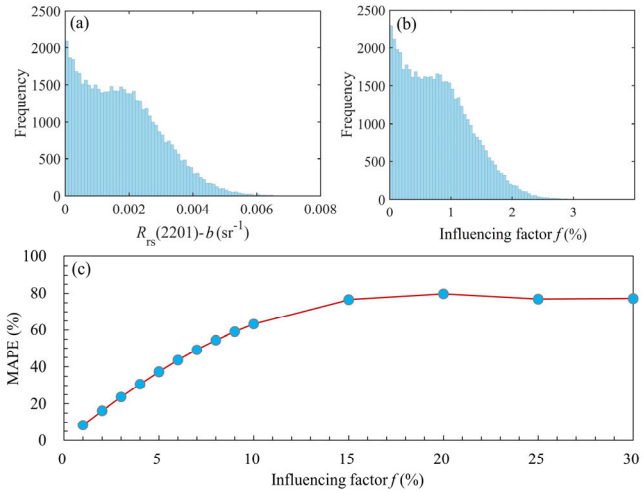


Fig. 7. (a) Frequency distributions of $R_{rs}(2201)$-b and (b) aquaculture facility influencing factor f within the mussel aquaculture ranching area. (c) MAPEs in Z_{sd} estimates corresponding to varying levels of the influencing factor f: 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, and 30%.

below 0.001 sr^{-1} [see Fig. 3(e)–(h)]. These findings show that, within the mussel aquaculture ranching area, aquaculture facility influence varies considerably among pixels. Furthermore, the influencing factor f also exhibits a wide range of variation, with values reaching up to 3% [see Fig. 7(b)], while most pixels show an f of around 1%.

As noted above, the maximum value of the aquaculture facility influencing factor f in our study area is approximately 3%, with most values around 1%. However, these influences are not negligible, and ignoring them would substantially reduce the accuracy of subsequent water quality parameter retrieval. To illustrate this effect, we quantified the degree to which an aquaculture facility influencing factor f of approximately 1% can alter Z_{sd}. Specifically, the MAPE was calculated between Z_{sd} values derived from $R_{rs}(\lambda)$ before and after corrections for aquaculture facility impacts. For the four Landsat images used, the MAPE for Z_{sd} exceeded 6.8%, with higher values observed in January and August 2022, reaching 8.5% and 7.9%, respectively. Therefore, although the aquaculture facility influence factor f in this study area is only about 1%, the resulting impact on Z_{sd} estimation is significant, underscoring the necessity for correcting the influence of aquaculture facilities.

In the mussel aquaculture zone of our study area, the density of aquaculture buoys is relatively low, yielding an influence factor f of around 1%. In other marine ranching areas, the coverage of aquaculture facilities may be much higher. To assess the potential impact of increased aquaculture facility coverage on Z_{sd} errors, a simulation experiment was conducted to analyze varying levels of aquaculture facility influence (f = 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, and 30%). Using in situ measurements of the aquaculture facility and water reflectance, (8) was applied to simulate $R_{rs}(\lambda)$ under varying degrees of aquaculture facility influence. The resulting simulated $R_{rs}(\lambda)$ values were then employed to retrieve Z_{sd}. We found that when the influencing factor f was 1%, the MAPE between Z_{sd} values derived from $R_{rs}(\lambda)$ before and after corrections reached 8.3% [see Fig. 7(c)]. This outcome closely aligns with real satellite data, where f measures around 1% and the MAPE for Z_{sd} is about 8%. When f ranges from 1% to 10%, the MAPE rises rapidly, reaching 23.7% at 3%, 37.4% at 5%, and 63.2% at 10%. Beyond 10%, the MAPE gradually saturates, reaching a very high value of 76.6% at f of 15% and stabilizing around 77%. This substantial increase in Z_{sd} uncertainties highlights the critical importance of correcting $R_{rs}(\lambda)$ within the mussel aquaculture ranching area to maintain accurate water transparency estimates. Therefore, special care must be taken to address and correct the impact of aquaculture facilities when retrieving water quality parameters in complex marine ranching regions using satellite observations.

B. Implications and Suggestions

Our research demonstrates that aquaculture facilities in marine ranching areas exert a measurable influence on $R_{rs}(\lambda)$, and the correction method proposed in this study effectively enhances the accuracy of $R_{rs}(\lambda)$ measurements, thereby improving Z_{sd} retrievals. As $R_{rs}(\lambda)$ serves as the foundational dataset for deriving various water quality parameters, applying this correction to mitigate facility-induced influences ensures that the corrected $R_{rs}(\lambda)$ can be reliably used to retrieve water environmental parameters, such as water transparency, chlorophyll-a concentration, and total suspended particulate matter concentration. Furthermore, a notable strength of our correction methodology lies in its foundation on comparative spectral analysis between pixels within the aquaculture area (potentially influenced by agriculture facilities) and those outside (serving as control areas unaffected by facilities). Thus, this approach is independent of specific satellite sensor settings, making it highly adaptable. By calculating the impact correction factors, our method can be applied to $R_{rs}(\lambda)$ measurements from various satellite sensors. As a result, the $R_{rs}(\lambda)$ correction framework proposed in this study provides a robust scientific basis for the reliable application of ocean color remote sensing technologies in the complex environments of marine ranching areas.

It is important to emphasize that the correction method proposed in this study is specifically tailored to white buoys in mussel ranching areas. While it may not be directly applicable to other types of aquaculture facilities, it provides a valuable technical reference for future research. Building on the research framework introduced here, once the remote sensing reflectance endmember spectra for additional aquaculture facilities are obtained, this method can be adapted to correct for their impacts in various aquaculture environments. Meanwhile, although this study focuses on water transparency estimation using Landsat imagery as a case example, the corrected $R_{rs}(\lambda)$ data have broader applicability. In future research, it is essential to develop algorithms for additional water quality parameters, such as chlorophyll-a concentration and total suspended particulate matter, specifically tailored to the complex conditions of aquaculture areas. We also aim to adapt the method for other satellite sensors, including Sentinel-2. These efforts will enable comprehensive and rapid assessments of multiple water environmental factors, thereby

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

strengthening the capability for effective monitoring and management of marine ranching areas and ultimately contributing to the sustainable and healthy development of the aquaculture industry.

V. Conclusion

This study addresses a critical gap in remote sensing applications for marine ranching by systematically quantifying the spectral distortions caused by aquaculture facilities and proposing an effective correction method. The results demonstrate that these structures introduce significant errors in $R_{rs}(\lambda)$ measurements, particularly in the SWIR region, leading to substantial inaccuracies in water quality retrievals. By incorporating an aquaculture facility influence factor and a pixel-based dynamic correction approach, our method effectively reduces these distortions, improving the accuracy of satellite-derived water transparency estimates.

The validation through Z_{sd} retrievals highlights the efficacy of this correction, particularly in mitigating errors that increase exponentially with aquaculture facility coverage. These findings emphasize the necessity of accounting for aquaculture-induced reflectance distortions to ensure reliable water quality assessments in marine ranching environments. Beyond the specific case study, the proposed method offers a broadly applicable framework that can be adapted to various aquaculture settings and integrated with multiple satellite sensors, enhancing the monitoring capabilities of remote sensing in coastal and offshore aquaculture management. Future research could explore the applicability of this approach across different satellite sensors, refine the correction model for varying aquaculture facility types, and extend its use to other water quality parameters critical for marine ecosystem health and sustainable aquaculture management.

Acknowledgment

The authors acknowledge U.S. Geological Survey (USGS) for providing Landsat 8 Satellite Data. Special thanks are extended to the research group led by Prof. Zhongping Lee at Xiamen University, Xiamen, China, for providing an excel-based tool that significantly facilitated the processing of in situ $R_{rs}(\lambda)$ data.

References

[1] G. Krause et al., "A revolution without people? Closing the people–policy gap in aquaculture development," *Aquaculture*, vol. 447, pp. 44–55, Oct. 2015, doi: 10.1016/j.aquaculture.2015.02.009.

[2] B. S. Halpern et al., "Spatial and temporal changes in cumulative human impacts on the world's ocean," *Nature Commun.*, vol. 6, no. 1, p. 7615, Jul. 2015, doi: 10.1038/ncomms8615.

[3] M. Jiao et al., "Construction and influencing factors of an early warning system for marine ranching ecological security: Experience from China's coastal areas," *J. Environ. Manage.*, vol. 335, Jun. 2023, Art. no. 117515, doi: 10.1016/j.jenvman.2023.117515.

[4] Y.-W. Du and W.-M. Cao, "DEPD model for evaluating marine ranching ecological security and its application in Shandong, China," *Ocean Coastal Manage.*, vol. 224, Jun. 2022, Art. no. 106206, doi: 10.1016/j.ocecoaman.2022.106206.

[5] M. Qin and M. Sun, "Effects of marine ranching policies on the ecological efficiency of marine ranching—Based on 25 marine ranching in Shandong province," *Mar. Policy*, vol. 134, Dec. 2021, Art. no. 104788, doi: 10.1016/j.marpol.2021.104788.

[6] J. Shen, J. Lv, and R. Liu, "A study on compensation scheme design of blue carbon sink for marine ranching in China," *J. Ocean Univ. China*, vol. 19, no. 3, pp. 68–75, Mar. 2020, doi: 10.16497/j.cnki.1672-335X.202003007.

[7] Y. Wang, T. Guo, T. C. E. Cheng, and N. Wang, "Evolution of blue carbon trading of China's marine ranching under the blue carbon special subsidy mechanism," *Ocean Coastal Manage.*, vol. 222, May 2022, Art. no. 106123, doi: 10.1016/j.ocecoaman.2022.106123.

[8] T. Rodrigues, E. Alcântara, F. Watanabe, and N. Imai, "Retrieval of Secchi disk depth from a reservoir using a semi-analytical scheme," *Remote Sens. Environ.*, vol. 198, pp. 213–228, Sep. 2017, doi: 10.1016/j.rse.2017.06.018.

[9] E. D. R. Maúre, G. Terauchi, J. Ishizaka, N. Clinton, and M. DeWitt, "Globally consistent assessment of coastal eutrophication," *Nature Commun.*, vol. 12, no. 1, p. 6142, Oct. 2021, doi: 10.1038/s41467-021-26391-9.

[10] K. Shi et al., "Long-term remote monitoring of total suspended matter concentration in lake Taihu using 250m MODIS-Aqua data," *Remote Sens. Environ.*, vol. 164, pp. 43–56, Jul. 2015, doi: 10.1016/j.rse.2015.02.029.

[11] L. Feng, X. Hou, and Y. Zheng, "Monitoring and understanding the water transparency changes of fifty large lakes on the Yangtze Plain based on long-term MODIS observations," *Remote Sens. Environ.*, vol. 221, pp. 675–686, Feb. 2019, doi: 10.1016/j.rse.2018.12.007.

[12] M. Doron, M. Babin, O. Hembise, A. Mangin, and P. Garnesson, "Ocean transparency from space: Validation of algorithms estimating Secchi depth using MERIS, MODIS and SeaWiFS data," *Remote Sens. Environ.*, vol. 115, no. 12, pp. 2986–3001, Dec. 2011, doi: 10.1016/j.rse.2011.05.019.

[13] W. Kim, J.-E. Moon, Y.-J. Park, and J. Ishizaka, "Evaluation of chlorophyll retrievals from geostationary ocean color imager (GOCI) for the North-East Asian region," *Remote Sens. Environ.*, vol. 184, pp. 482–495, Oct. 2016, doi: 10.1016/j.rse.2016.07.031.

[14] Y. Mao, S. Wang, Z. Qiu, D. Sun, and M. Bilal, "Variations of transparency derived from GOCI in the Bohai sea and the yellow sea," *Opt. Exp.*, vol. 26, no. 9, pp. 12191–12209, Apr. 2018, doi: 10.1364/oe.26.012191.

[15] C. Hu, Z. Lee, and B. Franz, "Chlorophyll aalgorithms for oligotrophic oceans: A novel approach based on three-band reflectance difference," *J. Geophys. Res., Oceans*, vol. 117, no. C1, Jan. 2012, Art. no. C01011, doi: 10.1029/2011jc007395.

[16] D. Blondeau-Patissier, J. F. R. Gower, A. G. Dekker, S. R. Phinn, and V. E. Brando, "A review of ocean color remote sensing methods and statistical techniques for the detection, mapping and analysis of phytoplankton Blooms in coastal and open oceans," *Prog. Oceanogr.*, vol. 123, pp. 123–144, Apr. 2014, doi: 10.1016/j.pocean.2013.12.008.

[17] J. E. O'Reilly and P. J. Werdell, "Chlorophyll algorithms for ocean color sensors–OC4, OC5 & OC6," *Remote Sens. Environ.*, vol. 229, pp. 32–47, Aug. 2019, doi: 10.1016/j.rse.2019.04.021.

[18] Z. Lee, K. L. Carder, and R. A. Arnone, "Deriving inherent optical properties from water color: A multiband quasi-analytical algorithm for optically deep waters," *Appl. Opt.*, vol. 41, no. 27, pp. 5755–5772, Sep. 2002, doi: 10.1364/ao.41.005755.

[19] P. J. Werdell et al., "An overview of approaches and challenges for retrieving marine inherent optical properties from ocean color remote sensing," *Prog. Oceanogr.*, vol. 160, pp. 186–212, Jan. 2018, doi: 10.1016/j.pocean.2018.01.001.

[20] D. S. F. Jorge et al., "A three-step semi analytical algorithm (3SAA) for estimating inherent optical properties over oceanic, coastal, and inland waters from remote sensing reflectance," *Remote Sens. Environ.*, vol. 263, Sep. 2021, Art. no. 112537, doi: 10.1016/j.rse.2021.112537.

[21] C. Hu, L. Feng, and Q. Guan, "A machine learning approach to estimate surface chlorophyll a concentrations in global oceans from satellite measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4590–4607, Jun. 2021, doi: 10.1109/TGRS.2020.3016473.

[22] S. Kolluru and S. P. Tiwari, "Modeling ocean surface chlorophyll–A concentration from ocean color remote sensing reflectance in global waters using machine learning," *Sci. Total Environ.*, vol. 844, Oct. 2022, Art. no. 157191, doi: 10.1016/j.scitotenv.2022.157191.

[23] C. R. McClain, "A decade of satellite ocean color observations," *Annu. Rev. Mar. Sci.*, vol. 1, no. 1, pp. 19–42, Jan. 2009, doi: 10.1146/annurev.marine.010908.163650.

[24] S. Groom et al., "Satellite ocean colour: Current status and future perspective," *Frontiers Mar. Sci.*, vol. 6, p. 485, Aug. 2019, doi: 10.3389/fmars.2019.00485.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.

[25] M. Bonansea, M. C. Rodriguez, L. Pinotti, and S. Ferrero, "Using multi-temporal Landsat imagery and linear mixed models for assessing water quality parameters in Río tercero reservoir (Argentina)," *Remote Sens. Environ.*, vol. 158, pp. 28–41, Mar. 2015, doi: 10.1016/j.rse.2014.10.032.

[26] Z. Lee, S. Shang, L. Qi, J. Yan, and G. Lin, "A semi-analytical scheme to estimate Secchi-disk depth from Landsat-8 measurements," *Remote Sens. Environ.*, vol. 177, pp. 101–106, May 2016, doi: 10.1016/j.rse.2016.02.033.

[27] G. Liu et al., "An OLCI-based algorithm for semi-empirically partitioning absorption coefficient and estimating chlorophyll a concentration in various turbid case-2 waters," *Remote Sens. Environ.*, vol. 239, Mar. 2020, Art. no. 111648, doi: 10.1016/j.rse.2020.111648.

[28] J. Tavora et al., "Recipes for the derivation of water quality parameters using the high-spatial-resolution data from sensors on board Sentinel-2A, Sentinel-2B, Landsat-5, Landsat-7, Landsat-8, and Landsat-9 satellites," *J. Remote Sens.*, vol. 3, p. 49, Jun. 2023, doi: 10.34133/remotesensing.0049.

[29] D. Liu et al., "Substantial increase of organic carbon storage in Chinese lakes," *Nature Commun.*, vol. 15, no. 1, p. 8049, Sep. 2024, doi: 10.1038/s41467-024-52387-2.

[30] W. Zhong, J. Lin, Q. Zou, Y. Wen, W. Yang, and G. Yang, "Hydrodynamic effects of large-scale suspended mussel farms: Field observations and numerical simulations," *Frontiers Mar. Sci.*, vol. 9, Aug. 2022, Art. no. 973155, doi: 10.3389/fmars.2022.973155.

[31] L. Cai, J. Yin, X. Yan, Y. Zhou, R. Tang, and M. Yu, "The environmental analysis and site selection of mussel and large yellow croaker aquaculture areas based on high resolution remote sensing," *Acta Oceanolog. Sinica*, vol. 43, no. 3, pp. 66–86, Mar. 2024, doi: 10.1007/s13131-023-2284-5.

[32] Z. Lee, Y.-H. Ahn, C. Mobley, and R. Arnone, "Removal of surface-reflected light for the measurement of remote-sensing reflectance from an above-surface platform," *Opt. Exp.*, vol. 18, no. 25, pp. 26313–26324, Dec. 2010.

[33] Q. Vanhellemont, "Adaptation of the dark spectrum fitting atmospheric correction for aquatic applications of the Landsat and Sentinel-2 archives," *Remote Sens. Environ.*, vol. 225, pp. 175–192, May 2019, doi: 10.1016/j.rse.2019.03.010.

[34] Q. Vanhellemont, "Sensitivity analysis of the dark spectrum fitting atmospheric correction for metre- and decametre-scale satellite imagery using autonomous hyperspectral radiometry," *Opt. Exp.*, vol. 28, no. 20, pp. 29948–29965, Sep. 2020, doi: 10.1364/OE.397456.

[35] J. Chen, S. Yin, R. Xiao, Q. Xu, and C. Lin, "Deriving remote sensing reflectance from turbid case II waters using green-shortwave infrared bands based model," *Adv. Space Res.*, vol. 53, no. 8, pp. 1229–1238, Apr. 2014, doi: 10.1016/j.asr.2014.01.014.

[36] A. A. Gitelson, J. F. Schalles, and C. M. Hladik, "Remote chlorophyll—A retrieval in turbid, productive estuaries: Chesapeake bay case study," *Remote Sens. Environ.*, vol. 109, no. 4, pp. 464–472, Aug. 2007, doi: 10.1016/j.rse.2007.01.016.

[37] J. R. Irons, J. L. Dwyer, and J. A. Barsi, "The next Landsat satellite: The Landsat data continuity mission," *Remote Sens. Environ.*, vol. 122, pp. 11–21, Jul. 2012, doi: 10.1016/j.rse.2011.08.026.

[38] L. Goddijn-Murphy, S. Peters, E. van Sebille, N. A. James, and S. Gibb, "Concept for a hyperspectral remote sensing algorithm for floating marine macro plastics," *Mar. Pollut. Bull.*, vol. 126, pp. 255–262, Jan. 2018.

[39] S. Wang et al., "Unveiling reflectance spectral characteristics of floating plastics across varying coverages: Insights and retrieval model," *Opt. Exp.*, vol. 32, no. 13, pp. 22078–22094, Jun. 2024.

[40] C. E. A. Carver and A. L. Mallet, "Estimating the carrying capacity of a coastal inlet for mussel culture," *Aquaculture*, vol. 88, no. 1, pp. 39–53, Jul. 1990, doi: 10.1016/0044-8486(90)90317-g.

[41] P. Duarte, M. J. Fernández-Reiriz, and U. Labarta, "Modelling mussel growth in ecosystems with low suspended matter loads using a dynamic energy budget approach," *J. Sea Res.*, vol. 67, no. 1, pp. 44–57, Jan. 2012, doi: 10.1016/j.seares.2011.09.002.

[42] A. C. Smaal, T. Schellekens, M. R. van Stralen, and J. C. Kromkamp, "Decrease of the carrying capacity of the oosterschelde estuary (SW delta, NL) for bivalve filter feeders due to overgrazing?" *Aquaculture*, vols. 404–405, pp. 28–34, Aug. 2013, doi: 10.1016/j.aquaculture.2013.04.008.

[43] D. R. Plew, C. L. Stevens, R. H. Spigel, and N. D. Hartstein, "Hydrodynamic implications of large offshore mussel farms," *IEEE J. Ocean. Eng.*, vol. 30, no. 1, pp. 95–108, Jan. 2005, doi: 10.1109/JOE.2004.841387.

[44] B. Dahlbäck and L. Å. H. Gunnarsson, "Sedimentation and sulfate reduction under a mussel culture," *Mar. Biol.*, vol. 63, no. 3, pp. 269–275, Jul. 1981, doi: 10.1007/BF00395996.

[45] A. Hatcher, J. Grant, and B. Schofield, "Effects of suspended mussel culture (Mytilus spp.) on sedimentation, benthic respiration and sediment nutrient dynamics in a coastal bay," *Mar. Ecol. Prog. Ser.*, vol. 115, pp. 219–235, Dec. 1994, doi: 10.3354/meps115219.

[46] J. Grant et al., "A model of aquaculture biodeposition for multiple estuaries and field validation at blue mussel (Mytilus edulis) culture sites in eastern Canada," *Can. J. Fisheries Aquatic Sci.*, vol. 62, no. 6, pp. 1271–1285, Jun. 2005, doi: 10.1139/f05-033.



Shengqiang Wang received the B.S. degree in physics from Beijing Normal University, Beijing, China, in 2008, and the Ph.D. degree in Earth and environmental sciences from Nagoya University, Nagoya, Japan, in 2014.

He is currently a Professor with the School of Marine Sciences, Nanjing University of Information Science and Technology, Nanjing, China, where he has been working on the ocean optical remote sensing, marine bio-optics, and satellite oceanography fields, since 2015.



Jiayu Meng received the M.S. degree in marine science from Nanjing University of Information Science and Technology, Nanjing, China, in 2025.

Since 2022, she has been working in the field of remote sensing of water quality parameters in marine ranching areas, with a particular focus on the impact of aquaculture facilities on retrieval accuracy and the development of correction methods.



Deyong Sun received the B.S. degree in geomatics engineering from Nanjing University of Technology, Nanjing, China, in 2005, and the Ph.D. degree in remote sensing technology and applications from Nanjing Normal University, Nanjing, in 2010.

He is currently a Professor with the School of Marine Sciences, Nanjing University of Information Science and Technology. His research focuses on ocean color remote sensing, especially coastal water optical properties, marine phytoplankton remote sensing, and marine disaster phenomenon monitoring.



Xin Zhang received the Ph.D. degree in cartography and geographic information systems from the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing, China, in 2004.

His research focuses on spatiotemporal analysis of remote sensing big data, with an emphasis on the theoretical methods and technological applications in aerospace remote sensing and geographic information science.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.



Zishen Li received the B.Sc. degree in surveying engineering from China University of Geosciences, Wuhan, China, in 2007, and the Ph.D. degree in geodesy and surveying engineering from the Institute of Geodesy and Geophysics, Chinese Academy of Sciences, Wuhan, in 2013.

His research focuses on high-precision positioning and navigation using the BeiDou Navigation Satellite System, with particular emphasis on reliable and accurate positioning, ionospheric monitoring and modeling, and autonomous navigation applications.



Xiumei Zhang received the Ph.D. degree in fisheries science from Tokyo University of Fisheries, Tokyo, Japan, in 1993.

She is currently a Professor with Zhejiang Ocean University, Zhoushan, China. Her research focuses on fish behavioral ecology and stock enhancement ecology, with particular emphasis on swimming behavior, reproductive behavior, and territoriality in fish.



Shuyan Lang received the M.S. degree in electromagnetic field and microwave technology from the University of Chinese Academy of Sciences, Beijing, China, in 2008.

Since 2008, she has been with the National Satellite Ocean Application Service, Beijing. Her primary research interests lie in satellite data processing and quality control, as well as the application of remote sensing products in oceanographic studies and climate change research.



Yongjun Jia received the Ph.D. degree in physical oceanography from the Institute of Oceanology, Chinese Academy of Sciences, Beijing, China, in 2010.

He was the Deputy Chief Designer of the ground application system for the HY-2 satellites and new generational ocean dynamic satellites, where he organized the framework and managed the construction of the ground application system for Chinese ocean dynamic satellites. His research interests include ocean satellite data processing, remote sensing of the ocean, and associated applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:19:13 UTC from IEEE Xplore. Restrictions apply.